Certified unconditional 67.3192911473% lower bound for simple zeros of the Riemann zeta function on the critical line

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Abstract

We give a reproducible proof record for

$$\liminf_{T\to\infty} \frac{N_0^s(T,2T)}{N(T,2T)} \geq 0.6731929114731422535099843283\ldots.$$

The argument uses the nonnegative cosine window $v(s) = \cos(1.47s)$ on $[-1/2, 1/2]$, jointly optimizes the window and the Gram-overlap correction, and proves a seven-point inequality by interval arithmetic. The certificate uses directed MPFR endpoint bounds, a 43,247-cell kernel table, a CWD2 derivative table, and exhaustive subdivision of 64 initial boxes. Every box reruns with `verified=true`. The stored table has SHA-256

`13213b84960fa629db0eac3ed78911480`
`66313cba84f4fa151cfcce749d8fc2c`.

Here "unconditional" means that no Riemann-hypothesis assumption or other unproved hypothesis about zeta zeros is made. The repository rerun passed locally; it has not received an external independent audit or peer review.

Verification status. The bound evaluation, byte-for-byte table regeneration, C++20 compilation, and all 64 boxes passed on the recorded local toolchain. This is an in-repository rerun of the supplied artifacts, not an external audit or peer review.

Contents

1 Statement and scope

Let $N(T,2T)$ count nontrivial zeros of the Riemann zeta function in $(T,2T]$, with multiplicity, and let $N_0^s(T,2T)$ count simple zeros on the critical line $\Re s = 1/2$. The main result is:

Theorem 1.1 (Certified simple-zero proportion)**.** *Without assuming the Riemann hypothesis (or any other conjecture about the location or simplicity of zeta zeros),*

$$\liminf_{T\to\infty} \frac{N_0^s(T,2T)}{N(T,2T)} \geq 0.67319291147314225350998432 83\ldots. \tag{1}$$

Equivalently, the certified percentage is $67.3192911473142253509\ldots\%$.

Here "unconditional" has its standard analytic-number-theory meaning: the inequality uses stated analytic theorems and explicit finite inequalities, but does not assume RH, a zero-density conjecture, pair correlation, or a model for the zeros. The general-window analytic input is imported, with provenance and a pinned commit in Section 8; this archive does not claim to have reproved that imported theory. The arithmetic constant, kernel table, and finite interval certificate are independently rerun here.

This paper is a compact guide to the repository. The current source note is `BELLMAN_COBBOUNDARY_PROOF.md`; `NEXT_FRONTIER.md` records the separate exploratory route. The executable protocol is in `tools/`, and the verification record is in `docs/verification.md`.

2 Window functional and joint choice

The imported general-window formulation associates to a nonnegative profile v on $[-1/2,1/2]$ the scale-free constant

$$c_1(v) = \frac{\left(\int_{-1/2}^{1/2} v(s)\,\mathrm{d}s\right)^2}{\int_{-1/2}^{1/2} v(s)^2\,\mathrm{d}s + \iint_{[-1/2,1/2]^2} |s-t|v(s)v(t)\,\mathrm{d}s\,\mathrm{d}t}, \qquad H(v) = 2 - \frac{1}{c_1(v)}. \tag{2}$$

The source for this nonnegative-window extension is Anthropic's official PDF, Section 7.1, equations (7.1)–(7.3), cited in Section 8.

We choose

$$v_\alpha(s) = \cos(\alpha s), \qquad \alpha = \frac{147}{100} = 1.47, \qquad |s| \leq \frac{1}{2}. \tag{3}$$

Put

$$I_0 = \int_{-1/2}^{1/2} \cos(\alpha s)\,\mathrm{d}s = \frac{2\sin(\alpha/2)}{\alpha}, \tag{4}$$

$$I_2 = \int_{-1/2}^{1/2} \cos^2(\alpha s)\,\mathrm{d}s = \frac{1}{2} + \frac{\sin\alpha}{2\alpha}. \tag{5}$$

For $(Tv)(s) = \int_{-1/2}^{1/2} |s-t|v(t)\,\mathrm{d}t$, one has $(Tv)'' = 2v$. Evenness and the endpoint derivative give

$$(Tv_\alpha)(s) = -\frac{2}{\alpha^2}\cos(\alpha s) + \frac{\sin(\alpha/2)}{\alpha} + \frac{2\cos(\alpha/2)}{\alpha^2}. \tag{6}$$

Thus, with $J_\alpha = \iint |s-t|v_\alpha(s)v_\alpha(t)\,\mathrm{d}s\,\mathrm{d}t$,

$$J_\alpha = -\frac{2I_2}{\alpha^2} + \left(\frac{\sin(\alpha/2)}{\alpha} + \frac{2\cos(\alpha/2)}{\alpha^2}\right) I_0. \tag{7}$$

At $\alpha = 1.47$,

$$I_0 = 0.91236347705837245140099940575\ldots, \qquad I_2 = 0.83840964278149010131290083 77\ldots,$$
$$J_\alpha = 0.26664517200057776931779678 55\ldots, \quad c_1(v_\alpha) = 0.75327223874790820890723816 32\ldots,$$
$$H_\alpha = 0.67245870940072934017051068 78\ldots.$$

The window alone is slightly below the best two-trace cosine choice; the gain comes from optimizing it jointly with the overlap correction.

3 Overlap kernel and finite certificate

The normalized limiting overlap kernel is

$$k_\alpha(x) = \frac{\int_{-1/2}^{1/2} \cos(\alpha s)\cos(2\pi x s)\,\mathrm{d}s}{\int_{-1/2}^{1/2}\cos(\alpha s)\,\mathrm{d}s} = \frac{\operatorname{sinc}(\pi x - \alpha/2) + \operatorname{sinc}(\pi x + \alpha/2)}{2\operatorname{sinc}(\alpha/2)}, \qquad w_\alpha(x) = k_\alpha(x)^2. \tag{8}$$

For six consecutive gaps $g_1, \ldots, g_6 \geq 0$, start from the seven-point functional F_0 and add the explicit five-gap coboundary

$$U(g_1, \ldots, g_5) = \frac{54g_1 - 123g_2 + 123g_4 - 54g_5}{1920000} \tag{9}$$
$$+ \frac{5971}{300000}(w(g_1) + w(g_2) - w(g_4) - w(g_5)), \tag{10}$$
$$F_B(g_1, \ldots, g_6) = F_0(g_1, \ldots, g_6) + U(g_2, \ldots, g_6) - U(g_1, \ldots, g_5). \tag{11}$$

The resulting pressure and nearest-neighbour coefficients are

$$(p_1, \ldots, p_6) = \frac{(946, 1177, 877, 877, 1177, 946)}{1920000},$$
$$(q_1, \ldots, q_6) = \left(\frac{31343}{100000}, \frac{1}{3}, \frac{105971}{300000}, \frac{105971}{300000}, \frac{1}{3}, \frac{31343}{100000}\right). \tag{12}$$

Here $\sum p_i = 1/320$, $\sum q_i = 2$, so the coboundary telescopes while preserving global pair-energy accounting. Its directed-MPFR CWD2 certificate proves

$$\mathcal{F}_B(g_1, \ldots, g_6) \geq \frac{577}{100000}, \qquad \alpha = \frac{147}{100}. \tag{13}$$

This is the sole new computer-assisted assertion.

The generator evaluates directed 256-bit MPFR enclosures at a grid of 4000 points. The kernel table is the unchanged 43,247-cell “CWK2“ file; the independently reimplemented derivative table is a big-endian “CWD2“ file.

The one-body pressure test leaves two cell components and their six-fold Cartesian product gives 64 initial boxes. The C++20 GMP verifier recursively splits the widest coordinate; pressure, interval, or tangent lower bounds prune each branch. The recorded aggregate is:

initial boxes	64
visited nodes	1,126,636
splits	563,286
pressure-pruned leaves	3,477
interval-pruned leaves	318,922
tangent-pruned leaves	240,951
maximum depth	55
unresolved terminal cells	0

Every invocation for box codes $0, 1, \ldots, 63$ returned `verified=true`. The certificate records target 577/100000 and the local derivative-table SHA-256

```
035946b4368fbeab578720109039bb4098
77f2c3728b672a1c1daff6c3e6f375.
```

At each tangent-pruned box the verifier encloses the Hessian with directed binary64 intervals, then converts every endpoint to an exact dyadic GMP rational and performs an exact-positive-definite LDL test. The tangent at the box midpoint is consequently a certified lower bound. The independent 64-box run satisfies the exact tree identity

$$1{,}126{,}636 - 563{,}286 = 3{,}477 + 318{,}922 + 240{,}951 = 563{,}350,$$

with no unresolved terminal cells.

4 From local gaps to a Gram defect

Let $y_1 < \cdots < y_m$ be normalized ordinates and $g_i = y_{i+1} - y_i$. Define

$$\begin{aligned} &E_m = 2 \sum_{1 \le i < j \le m} w_\alpha(y_j - y_i), \\ &P_m = \text{the nonnegative weighted gap sum induced by the } p_i \text{ in (12).} \end{aligned} \tag{14}$$

Summing (13) over consecutive windows gives

$$E_m + P_m \ge \frac{577}{100000}(m-6). \tag{15}$$

Across a canonical block, the total pressure is $(m-6)\sum_i p_i = (m-6)/320$; after averaging at $m = 183$ this is the tax 59/19520 used below.

The stability-enhanced rank–trace input is

$$S \ge H_\alpha N + \mathcal{D}(M^\circ) - o(N), \qquad \mathcal{D}(G) = \operatorname{tr}\Psi(G), \tag{16}$$

where

$$\Psi(t) = \begin{cases} (t-1)^2, & 0 \le t \le 2, \\ 2t-3, & t \ge 2. \end{cases} \tag{17}$$

The imported upstream commit contains the coarser Gram-defect lemma. The finite trace–energy implication used here is proved directly in `docs/trace_energy_envelope.md`; its short derivation follows.

For an $m \times m$ positive-semidefinite unit-diagonal Gram matrix, let λ_i be its eigenvalues, $x_i = \lambda_i - 1$, and $E = \sum_i x_i^2 = \operatorname{tr}(G-I)^2$. Let $L = \{i : x_i > 1\}$, $k = |L|$, $R = \sum_L x_i$, and $Q = \sum_L x_i^2$. Since the linear branch saves $(x_i - 1)^2$ on L,

$$\mathcal{D}(G) = E + 2R - k - Q. \tag{18}$$

If $k \ge 2$, Cauchy–Schwarz on the other $m-k$ coordinates gives $E - Q \ge R^2/(m-k)$ and hence

$$\mathcal{D}(G) \ge 2R - k + \frac{R^2}{m-k} \ge \frac{km}{m-k} > 2. \tag{19}$$

If $k = 0$, then $\mathcal{D}(G) = E$. If $k = 1$, writing the large coordinate as $r > 1$, Cauchy gives $r \le \sqrt{(m-1)E/m}$ and

$$\mathcal{D}(G) \ge E + 2r - 1 - r^2. \tag{20}$$

The correction $2r - r^2$ decreases for $r \geq 1$. For the value $A = 577(183-6)/100000 = 1.02129$, these cases imply

$$E + P \geq A, \quad P \geq 0 \Longrightarrow \quad \mathcal{D}(G) + P \geq \Phi_m(A), \tag{21}$$

where

$$\Phi_m(E) = \begin{cases} E, & 0 \leq E \leq m/(m-1), \\ 2\sqrt{(m-1)E/m} - 1 + E/m, & E \geq m/(m-1). \end{cases} \tag{22}$$

This is the only trace–energy range used below; no global minimizer claim is needed.

5 The certified constant

Choose $m = 183$ and

$$A = \frac{577}{100000}(183-6) = \frac{102129}{100000} = 1.02129. \tag{23}$$

The second branch of (22) gives

$$B := \Phi_{183}(A) = 2\sqrt{\frac{18587478}{18300000}} - 1 + \frac{102129}{18300000} \tag{24}$$

$$= 1.02122878529298216614894017 66\ldots. \tag{25}$$

The averaged pressure loss is

$$\frac{59}{19520} = 0.0030225409836065573770491803\ldots. \tag{26}$$

Pinching to full blocks for each shift and averaging gives

$$\mathcal{D}(M^{\circ}) \geq \frac{B}{183} S - \frac{59}{19520} N - o(N). \tag{27}$$

Substitution into (16) yields

$$\frac{H_{\alpha} - 59/19520}{1 - B/183} = 0.6731929114731422535099843283\ldots. \tag{28}$$

This proves Theorem 1.1; `compute_joint_bound.py` prints the same value to 90 digits.

6 Global spectral dual and future route

The archive also records a global route that is not needed for the theorem. For $t \geq 0$, the scalar identity

$$\Psi(t) = \sup_{h \leq 2} \left\{ h(t-1) - \frac{h^2}{4} \right\} \tag{29}$$

implies by spectral calculus

$$\operatorname{tr} \Psi(M) = \sup_{H = H^*, H \preceq 2I} \left[\operatorname{tr} H(M - I) - \frac{1}{4} \operatorname{tr} H^2 \right]. \tag{30}$$

Choosing a connection Laplacian $L(q) \succeq 0$ from nonnegative edge weights and setting $H = S - L(q)$ gives a global capacitated-edge certificate. The hard-capacity condition $\sum_j q_{ij} \leq 2$ removes the diagonal penalty and allows edges across old block boundaries.

The finite Bellman-coboundary certificate above is part of the strict bound. The separate global spectral-dual construction remains a future route for removing block-boundary pressure loss; the supplied double-precision reconnaissance for that route is not interval certified and is not used here.

7 Reproducibility and trust boundary

From the repository root, the Windows entry point is:

```
powershell -ExecutionPolicy Bypass -File scripts/verify.ps1
```

On a Unix-like system with Python 3, MPFR, and a C++20 compiler, the corresponding core steps are:

```
python3 tools/compute_joint_bound.py
python3 tools/run_generate_joint_kernel_table.py \
  -output /tmp/cos147-kernel.bin
cmp /tmp/cos147-kernel.bin data/cos147-kernel.bin
python3 tools/generate_coboundary_derivative_table.py \
  -output /tmp/cos147-derivatives.bin
cmp /tmp/cos147-derivatives.bin data/cos147-derivatives.bin
g++ -O2 -std=c++20 -frounding-math -ffp-contract=off \
  tools/verify_coboundary.cpp -lgmp -o /tmp/verify_coboundary
```

Run the resulting verifier with the stored table for box codes 0 through 63. The table must be byte-identical. The local run used Python 3.12.12, Strawberry MinGW g++ 13.2.0, and MPFR 4.2.1 through its Windows DLL; the local derivative table was generated with that 4.2.1 toolchain (the unavailable source package reported MPFR 4.2.2). No WSL distribution or Clang fallback was available or required.

The trust boundary is explicit: the local run checks the arithmetic constant, table generation, source compilation, and all finite boxes. It does not replace an independent implementation, formal proof checking, or peer review of the imported analytic framework and asymptotic passage. The numerical reconnaissance file is explicitly non-rigorous.

8 Provenance and repository

The code, certificates, proof notes, and source table are archived at

`https://github.com/tawanerguo-cn/zeta-simple-zeros`

with Zenodo concept DOI `https://doi.org/10.5281/zenodo.21890630`.

The earlier Gram-defect framework is imported from the pinned `ainta/zeta-simple-zeros` snapshot, at commit

`040c5e899e658aed7b56a2a87f501798fe10761d`.

That commit supplies the coarser $\min(1, E)$ lemma; it is not cited as a proof of the sharper finite implication in Section 4. The nonnegative-window analytic extension is recorded in Anthropic's official PDF, Section 7.1, equations (7.1)–(7.3), official PDF; the checked copy has SHA-256

`6792988e6cd0e17690621ce898abd5d5`
`34f98407741bc7cb14bbe7d07c77d72e`.

References

[1] ainta, *zeta-simple-zeros*, repository snapshot at commit `040c5e899e658aed7b56a2a87f501798fe10761d`, `https://github.com/ainta/zeta-simple-zeros/tree/040c5e899e658aed7b56a2a87f501798fe10761d`.

[2] Anthropic, official Claude mathematical paper, Section 7.1, equations (7.1)–(7.3), `https://www-cdn.anthropic.com/564f962e60643842f5fcb4a17c9dbc8f608f1c37.pdf`.

[3] tawanerguo-cn, *zeta-simple-zeros: certified unconditional 67.3192911473% lower bound for simple zeros of the Riemann zeta function*, `https://github.com/tawanerguo-cn/zeta-simple-zeros`, Zenodo concept DOI `https://doi.org/10.5281/zenodo.21890630`.